Heavy-tailed Value at Risk Models

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1 Economical Description of the Model

The model we are developing is essentially a Value at Risk (VaR) operational risk capital model which quantifies a firm's operational risk capital via a key risk scenario approach. For each scenario the model use agreed inputs (average operating event, frequency per year, median loss amount, and severe 1 in 25 year loss amount) to fit frequency and severity distributions (a Poisson distribution is assumed for frequency and a {Lognormal, stable, etc. } distribution for severity). The model then convolves the fitted frequency and severity distributions to estimate a Value at Risk (VaR) at the 99.5 % confidence level (also referred to as the "1 in a 200 year" level), which is the capital estimation, for each operational risk category (also referred to as"units of measure" or "UOM"). Our intention is to also implement a diversification benefit/aggregation between scenarios using a Gaussian copula with marginal distributions corresponding to the convolved annual loss distributions of each UOM to estimate the aggregate operational risk loss at 99.5 % confidence.

2 VaR Models via Compound Loss Distributions

We want to quantify the amount of capital that must be added to the financial position X in order the make it "acceptable", i.e. we want to calculate the **risk** of X. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Define the Lebesgue space

$$L^1 := \{X : \Omega \to \mathbb{R} \,|\, \mathbb{E}[|X|] < \infty\} \tag{2.1}$$

of random variables whose expected values have a finite absolute value. The **Value at Risk** of $X \in L^1$ at level $\alpha \in (0, 1)$ is defined as

$$\mathrm{VaR}_\alpha(X) := \inf \{y \in \mathbb{R} \mid \mathbb{P}(X + y < 0) \leq \alpha\} . \tag{2.2}$$

Now, we want to calculate the **total loss** of X in order to quantify the required risk capital. We use the **collective model** from actuarial mathematics for non-life insurance which involves the random **loss frequency** $N(t)$ up to time t and positive i.i.d. **loss severities** $(S_k)_{k\in\mathbb{N}}$. The total loss of X at time $t \geq 0$ is given by

$$Y(t) := \sum_{k=1}^{N(t)} S_k = \sum_{k=0}^{\infty} S_k \cdot \mathbb{1}_{\{N(t)=k\}}. \tag{2.3}$$

We say that Y is **compound distributed**. One can show that its cumulative distribution function (CDF) has the form

$$\mathbb{P}(Y(t) \leq x) = \sum_{k=0}^{\infty} \mathbb{P}(S_k \leq x) \cdot \mathbb{P}(N(t) = k). \tag{2.4}$$

If then $\mathbb{E}[S_1],\ \mathbb{E}[N(t)] < \infty$, then **Wald's formula** hold, viz.

$$\mathbb{E}[Y(t)] = \mathbb{E}[N(t)] \cdot \mathbb{E}[S_1]. \tag{2.5}$$

If $\mathbb{E}[S_1^2],\ \mathbb{E}[N(t)^2] < \infty$, then the **Blackwell-Girshick identity**

$$\operatorname{Var}(Y(t)) = \mathbb{E}[N(t)] \cdot \operatorname{Var}(S_1) + \operatorname{Var}(N(t)) \cdot \mathbb{E}[S_1] \tag{2.6}$$

holds. In the default case where the loss frequency $N(t)$ is Poisson distributed, we say that Y is **compound Poisson** distributed.

2.1 Loss Frequency

The losses occur at the random ***n*-th jump times** $(T_n)_{n\in\mathbb{N}_0}$ with $T_0 := 0$. In the default case, the continuous **holding times** $(\Delta T_n)_{n\in\mathbb{N}} := (T_n - T_{n-1})_{n\in\mathbb{N}}$ are exponentially i.i.d., i.e. $\Delta T_1 \sim \operatorname{Exp}(\lambda)$ for some parameter $\lambda > 0$ with

$$f_{\Delta T_1}(x) = \left(\lambda e^{-\lambda x}\right) \cdot \mathbb{1}_{\{x\geq 0\}} \tag{2.7}$$

for every $x \in \mathbb{R}$. One can show that then the **loss frequency** up to time $t \geq 0$, i.e.

$$N(t) := \#\left\{n \in \mathbb{N} \mid T_n \leq t\right\} = \sup\{n \in \mathbb{N} : T_n \leq t\}, \tag{2.8}$$

is Poisson distributed with parameter λt, i.e. $N(t) \sim \Pi(\lambda t)$. Hence, in this case, $(N(t))_{t\geq 0}$ is called a **Poisson process** with jumps of height 1 being right-continuous. The exponential distribution has a unique property: It is the only continuous distribution which is memoryless, i.e. for $X \sim \mathsf{Exp}(\lambda)$ one has

$$\mathbb{P}(X > t + s \mid X > s) = \mathbb{P}(X > t) \tag{2.9}$$

for every $s, t \geq 0$. Hence, by inheriting the memorylessness of the exponential distribution, the Poisson process has the Markov property in particular.

In more general, the distribution of the loss frequency $N(t)$ could be chosen from the **Panjer class** which consists of exactly 3 discrete distributions:

- The **Poisson** distribution $\Pi(\lambda)$, with $\lambda > 0$,
- The **binomial** distribution $\mathsf{B}(n, p)$, with $n \in \mathbb{N}$, $p \in (0, 1)$,
- The **negative binomial** distribution $\mathsf{B}^-(\nu, p)$, with $\nu > 0$, $p \in (0, 1)$.

Both the binomial and the negative binomial distribution also have interpretations with respect to the Poisson Process $(N_t)_{t\geq 0}$. Let $0 < s < t$ and assume that $N(t) = n$ for some $n \in \mathbb{N}_0$. Then one can show that the following **conditional distribution** follows a binomial distribution, viz.

$$\mathcal{P}_{N(s)|N(t)=n} = \mathsf{B}\left(n, \frac{s}{t}\right). \tag{2.10}$$

We could also **stochastically scale** the time parameter t of the Poisson process $(N(t))_{t\geq 0}$ with $\lambda = 1$ using an independent Erlang distributed random variable $\Theta \sim \mathsf{Erl}(\varrho, n)$ for some $\varrho > 0$ and $n \in \mathbb{N}$. The Erlang distribution is the distribution of the sum of n independent exponential random variables with parameter $1/\varrho$ each. It has the PDF

$$f_{\varrho,n}(x) = \frac{\varrho^n}{(n-1)!} x^{n-1} e^{-\varrho x} \mathbb{1}_{(0,\infty)}(x). \tag{2.11}$$

Now one can show that

$$\mathcal{P}_{N(t\Theta)} = \mathsf{B}^- \left(n, \frac{\varrho}{t+\varrho} \right). \tag{2.12}$$

Together with the initial value $p_0 \in (0,1)$, only these three distributions fulfill the **Panjer recursion**

$$p_k := \mathbb{P}(N(t) = k) = \left(a + \frac{b}{k} \right) \cdot p_{k-1}, \quad p_0 \in (0,1) \tag{2.13}$$

for every $k \in \mathbb{N}$, resulting in the following table:

Distribution	$\mathbb{P}(N(t)=k)$	a	b	$\mathbb{E}[N]$	$\mathrm{Var}(N)$
$\Pi(\lambda t)$	$e^{-(\lambda t)} \frac{(\lambda t)^k}{k!}$	0	λt	λt	λt
$\mathsf{B}(n,p)$	$\binom{n}{k} p^k (1-p)^{n-k}$	$-\frac{p}{1-p}$	$-(n+1)a$	np	$np(1-p)$
$\mathsf{B}^-(\nu,p)$	$\binom{\nu+k-1}{k} p^\nu (1-p)^k$	$1-p$	$(\nu-1)a$	$\frac{\nu(1-p)}{p}$	$\frac{\nu(1-p)}{p^2}$

Hence in this case, we can implement an easy algorithm which generates the probability mass function (PMF) of the loss frequency distribution.

2.2 Loss Severity Distributions

The **loss severities** $(S_k)_{k\in\mathbb{N}}$ are modeled as i.i.d. positive random variables. They are defined on the same probability space as the n-th loss jump times $(T_n)_{n\in\mathbb{N}}$, but both are assumed to be independent. We can choose the severity distribution from all one-sided distributions with positive support. Which one suites the financial position best will be a main part of the following considerations.

If we choose the loss severities $(S_k)_{k\in\mathbb{N}}$ to be discrete i.i.d., then we often get an easy recursion algorithm in order to determine the PMF of $Y(t_0)$ at a fixed point in time $t_0 \geq 0$: If N possess any distribution of the Panjer class $\{\Pi(\lambda),\ \mathrm{B}(n,p),\ \mathrm{B}^-(\nu,p)\}$ with PMF $(q_k)_{k\in\mathbb{N}_0}$ and $q_0 \in (0,1)$ and if $(p_k)_{k\in\mathbb{N}_0}$ is the PMF of the total loss $Y(t_0)$, the **Panjer recursion** formula

$$p_k = \sum_{j=1}^{k} \left(a + \frac{b \cdot j}{k} \right) \cdot \mathbb{P}(S_1 = j) \cdot p_{k-j} \tag{2.14}$$

with initial value $p_0 = q_0 \in (0, 1)$ holds for every $k \in \mathbb{N}$ with a and b given in the table above.

Next, we classify feasible severity distributions with respect to their tail asymptotics as $x \to \infty$. They appear ordered from very light-tailed to very heavy-tailed.

Light-Tailed Distributions

We begin with **light-tailed** loss severity distributions. Their tails decay faster than any power law.

Super-Exponential Tails

Let X follow an ordinary centered normal distribution, i.e. $X \sim \mathcal{N}(0, \sigma^2)$, then $Y = |X|$ is **half-normal distributed**, i.e. $X \sim \mathsf{half}\text{-}\mathcal{N}(\sigma)$ for some parameter $\sigma > 0$. It has the PDF

$$f_Y(x; \sigma) = \frac{\sqrt{2}}{\sigma\sqrt{\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \cdot \mathbb{1}_{[0,\infty)]} \tag{2.15}$$

with tail asymptotics

$$\mathbb{P}(X > x) \sim \frac{\sigma}{x\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \tag{2.16}$$

as $x \to \infty$, i.e. its tail decays faster then exponential.

Exponential Tails

Let X be **exponential distributed**, i.e. $X \sim \mathsf{Exp}(\lambda)$, with parameter $\lambda > 0$ and PDF

$$f(x; \lambda) = \lambda e^{-\lambda x} \cdot \mathbb{1}_{[0,\infty)}(x). \tag{2.17}$$

Its tail decays exponentially, since

$$\mathbb{P}(X > x) \sim e^{-\lambda x} \tag{2.18}$$

as $x \to \infty$.

Sub-Exponential Tails

Now follows a distribution with a tail that decays somewhat more slowly than the exponential distribution: A random variable X is said to follow a **log-normal distribution**, denoted by $X \sim \ln\text{-}\mathcal{N}(\mu, \sigma^2)$, if and only if $\ln(X) \sim \mathcal{N}(\mu, \sigma^2)$ with mean μ and variance σ^2. This leads to the PDF

$$f(x; \mu, \sigma^2) = \frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\ln(x) - \mu)^2}{2\sigma^2}\right). \tag{2.19}$$

Since $\ln(x) \ll x$ as $x \to \infty$, the tail decays slower than the exponential $\exp(-x)$, but for every $\alpha > 0$ it still decreases faster than any power law $x^{-\alpha}$. Therefore, we classify its tail asymptotics as light, but sub-exponential.

Heavy-Tailed Distributions

Now, we treat PDFs whose tails decay according to a power law. We consider the Lomax distribution as the Pareto type II distribution with location parameter $\mu = 0$. We further treat the one-sided stable distributions with tail parameter $\alpha \in (0, 1)$ and skewness parameter $\beta = 1$. We choose the Lévy distribution – where $\alpha = 1/2$ – as the canonical example for this distribution family. Since financial data is modeled around a tail parameter $\alpha \in (1.6, 1.9)$, we set the Lévy distribution as the absolute bottom of this parameter scale and only consider the Lomax distributions where $\alpha > 1$.

Long Tails

Let X be **Lomax distributed**, i.e. $X \sim \mathsf{Lomax}(\alpha, \lambda)$, with tail parameter $\alpha > 1$ and scale parameter $\lambda > 0$. It has the PDF

$$f(x; \alpha, \lambda) = \frac{\alpha \lambda^\alpha}{(x + \lambda)^{\alpha+1}} \cdot \mathbb{1}_{[0,\infty)}(x). \tag{2.20}$$

Its tail decays polynomial, since

$$\mathbb{P}(X > x) \sim \lambda^\alpha x^{-\alpha} \tag{2.21}$$

as $x \to \infty$.

Fat Tails

Lévy distribution.

Exercises

1. Implement the Panjer recursion formula (2.14) for the loss frequency $N(t)$ in the Panjer class $\{\Pi(\lambda t), \mathsf{B}(n, p), \mathsf{B}^-(\nu, p)\}$. Assume that the loss severities are i.i.d. according to the following positively, discrete distributions:

   (i) Uniform distribution.
   (ii) Geometric distribution.
   (iii) Poisson distribution.
   (iv) Yule-Simon distribution (heavy-tailed).
   (v) Zeta distribution (heavy-tailed).
   (vi) Zipf-Mandelbrot law (heavy-tailed).

   Calculate the PMFs of the resulting compound distributions and plot them for comparison.

3 VaR Models with Return Distributions

3.1 Tail Classification

A priori, we can arbitrarily choose the severity distributions from the huge number of probability distributions. For our purpose, it is useful to categorize them with respect their tail asymptotics.

Light Tales

In 1900, Louis Bachelier modeled financial asset returns by **Gaussian distributed** random variables $X \sim \mathcal{N}(\mu, \sigma^2)$ with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 \in (0, \infty)$. We have the PDF

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{3.1}$$

and CDF

$$F(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-\frac{t^2}{2}} \, \mathrm{d}t.$$

In case of the **standard normal distribution** $\mathcal{N}(0,1)$, we define $\varphi(x) := f(x; 0, 1)$ and $\Phi(x) := F(x; 0, 1)$, respectively.

Returns were viewed as the sum of i.i.d. random variables with finite variance, each representing individual investment decisions. And indeed, from that perspective, the normality assumption is mathematically justified by the following fundamental **Central Limit Theorem** (CLT).

Theorem (CLT)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically Gaussian distributed random variables with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 \in (0, \infty)$. Define the **standardized**, i.e. centered and normalized sum variable S_n^* as*

$$S_n^* := \frac{S_n - n\mu}{\sqrt{n\sigma^2}} := \frac{\sum_{k=1}^n X_k - n\mu}{\sqrt{n\sigma^2}}.$$

Then

$$S_n^* \xrightarrow{d} \mathcal{N}(0,1) \quad (n \to \infty)$$

holds true.

But empirical observations show returns that are larger than those predicted by the normal distribution. Because of its exponentially decaying tails, i.e.

$$\mathbb{P}(X > x) \sim \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad (x \to \infty) \tag{3.2}$$

the normal distribution assigns very low probabilities to extreme events, effectively cutting them off. In this case, the distribution is said to have **light tails**.

Heavy Tails

We now consider distributions with **heavy tails**, i.e., those that exhibit power-law asymptotic behavior. We first consider the Pareto distribution, which is indeed often used to model loss severities in actuarial science.
Why did Bachelier's model not work out properly, i.e. why did the CLT fail? There are three possibilities:

- The returns may not originate from a finite-variance distribution,
- from non-identically distributed return variables,
- from dependencies between them

– or from a combination of all these factors. The second and third reasons are hard to handle, but for distributions of infinite variance there exists a generalization of the CLT. It states, that the limiting distribution of sums of such variables is stable (Nolan, 2010). This, together with the fact that stable distributions are leptokurtic and feature fat tails and asymmetry, gives us a modeling tool justified by reasoning. In 1963, Mandelbrot proposed stable laws as an alternative model for asset returns, although they were introduced much earlier in 1925 by Paul Lévy. They are called stable because the sum of two independent α-stable distributed random variables is again α-stable distributed. This is also true for the Gaußian distribution and indeed, the Gaussian distribution is stable with parameter $\alpha = 2$. There are only 3 (standard) stable distributions with analytically known PDFs and CDFs:

- The **Gaussian** distribution $\mathcal{N}(\mu = 0,\ \sigma^2 = 1)$ with $\mu \in \mathbb{R},\ \sigma^2 > 0$,
- The **Cauchy** distribution $\mathsf{Cau}(\gamma = 1,\ \delta = 0)$ with $\gamma > 0,\ \delta \in \mathbb{R}$,
- The **Lévy** distribution $\mathsf{Lévy}(\gamma = 1, \delta = 0)$ with $\gamma > 0,\ \delta \in \mathbb{R}$.

We get the following table of analytically known stable distributions:

Distribution	α	**PDF**	**CDF**	$\mathbb{P}(X > x)$
$\mathcal{N}(\mu, \sigma^2)$	2	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	$\mathcal{O}\left(\frac{1}{x}\exp\left(-\frac{x^2}{2}\right)\right)$
$\mathsf{Cau}(\gamma, \delta)$	1	$\frac{1}{\pi}\frac{\gamma}{(x-\delta)^2+\gamma^2}$	$\frac{1}{\pi}\arctan\left(\frac{x-\delta}{\gamma}\right) + \frac{1}{2}$	$\mathcal{O}\left(\frac{1}{x}\right)$
$\mathsf{Lévy}(\gamma, \delta)$	1/2	$\sqrt{\frac{\gamma}{2\pi}}\frac{e^{-\frac{\gamma}{2(x-\delta)}}}{(x-\delta)^{3/2}}$	$2 - 2\Phi\left(\sqrt{\frac{\gamma}{x-\delta}}\right)$	$\mathcal{O}\left(\frac{1}{\sqrt{x}}\right)$

According to [Misiorek, A., Weron, R. (2010). 2nd edition of Handbook of Computational Statistics: Concepts and Methods, J.E. Gentle, W. Härdle, Y. Mori (eds.), Springer-Verlag, forthcoming in 2011/2012. This version: 31.07.2010], typical parameters for financial data modeled by stable processes are in the range of $1.6 < \alpha < 1.9$. Hence, they cannot be expressed in a closed analytical form like the distributions above and more sophisticated methods must be invoked.

3.2 General Stable Distributions

Each of the stable distributions is uniquely characterized by four parameters:

- The **index of stability** $\alpha \in (0, 2]$ causes the rate of the tail decay.
- The **skewness** parameter $\beta \in [-1, 1]$ defines the asymmetry.
- The **scale** parameter $\gamma \geq 0$.
- The **location** parameter $\delta \in \mathbb{R}$.

In general, the stable distribution do not have closed expressions for their PDFs and CDFs, except for the three distributions above. But they can be explicitly described by their **characteristic function** (CF) $\varphi(\xi) = \mathbb{E}\left[e^{iX}\right](\xi)$. It is well-known that the stable PDF p and all their derivatives exist in $C_0^\infty \cap L^1(\mathbb{R})$ – we just cannot express it analytically. In contrast, [cf. Nolan, J. (2020). Univariate Stable Distributions. Models for Heavy Tailed Data. Springer Series in Operations Research and Financial Engineering.], the CF of a **standard stable** distributed random variable $Z \sim \mathsf{S}(\alpha, \beta) = \mathsf{S}(\alpha, \beta, 1, 0; 1)$ can be written as

$$\ln(\varphi(\xi)) = \begin{cases} -|\xi|^\alpha \left(1 - i\beta \operatorname{sign}(\xi) \tan\left(\frac{\pi\alpha}{2}\right)\right), & \alpha \neq 1, \\ -|\xi| \left(1 + i\beta \operatorname{sign}(\xi) \frac{2}{\pi} \ln|\xi|\right), & \alpha = 1. \end{cases} \tag{3.3}$$

We can add a scale parameter $\gamma \geq 0$ and a location parameter $\delta \in \mathbb{R}$ in different ways which leads to the following two most common parametrizations.

A stable random variable obeys the **0-parametrization**, $X \sim \mathsf{S}(\alpha, \beta, \gamma, \delta; 0)$, if

$$X \overset{d}{=} \begin{cases} \gamma(Z - \beta \tan\left(\frac{\pi\alpha}{2}\right)) + \delta, & \alpha \neq 1, \\ \gamma Z + \delta, & \alpha = 1, \end{cases}$$

where the standard stable distributed Z follows (3.3), cf. [Nolan, J. P. (1997). Numerical Calculation of Stable Densities and Distribution Functions, Communications in Statistics – Stochastic Models 13: 759–774.]. Then X has the CF

$$\ln(\varphi(\xi)) = \begin{cases} i\delta\xi - \gamma^\alpha |\xi|^\alpha \left(1 + i\beta \operatorname{sign}(\xi) \tan\left(\frac{\pi\alpha}{2}\right) \left(|\gamma\xi|^{1-\alpha} - 1\right)\right), & \alpha \neq 1, \\ i\delta\xi - \gamma|\xi| \left(1 + i\beta \operatorname{sign}(\xi) \frac{2}{\pi} \ln \gamma|\xi|\right), & \alpha = 1. \end{cases}$$

Here, the CF (and hence the PDF and CDF) are jointly continuous in all four parameters what is useful for numerical concerns.

In contrast, a stable random variable obeys the **1-parametrization**, $Y \sim \mathsf{S}(\alpha, \beta, \gamma, \delta; 1)$, if

$$Y \stackrel{d}{=} \begin{cases} \gamma Z + \delta, & \alpha \neq 1, \\ \gamma Z + \delta + \beta \frac{2}{\pi} \gamma \ln(\gamma), & \alpha = 1, \end{cases}$$

where the standard stable distributed Z follows (3.3). Then Y has the CF

$$\ln(\varphi(\xi)) = \begin{cases} i\delta\xi - \gamma^\alpha |\xi|^\alpha \left(1 - i\beta \operatorname{sign}(\xi) \tan\left(\frac{\pi\alpha}{2}\right)\right), & \alpha \neq 1, \\ i\delta\xi - \gamma|\xi| \left(1 + i\beta \operatorname{sign}(\xi) \frac{2}{\pi} \ln|\xi|\right), & \alpha = 1. \end{cases}$$

Furthermore, when $\alpha > 1$ the mean of the distribution exists and is equal to δ. In general, the p-th moment of a stable random variable is finite if and only if $p > \alpha$, i.e. for $X \sim \mathcal{S}^\alpha(\beta, \gamma, \delta)$ one has

$$\mathbb{E}\left[|X|^p\right] < \infty \quad \Leftrightarrow \quad p < \alpha. \tag{3.4}$$

Thus, when $\alpha < 2$ the variance is infinite and the tails exhibit a power-law behavior. More precisely, using a CLT-type argument it can be shown that (Janicki and Weron, Janicki, A. and Weron, A. (1994a). Can one see α-stable variables and processes, Statistical Science 9: 109–126.; Samorodnitsky and Taqqu, 1994, see above):

$$\lim_{x\to\infty} x^\alpha \mathbb{P}(X > x) = C_\alpha (1+\beta)\gamma^\alpha,$$
$$\lim_{x\to-\infty} x^\alpha \mathbb{P}(X < -x) = C_\alpha (1+\beta)\gamma^\alpha,$$

with the constant $C_\alpha = \left(2 \int_0^\infty t^{-\alpha} \sin(t)\,\mathrm{d}t\right)^{-1} = \frac{1}{\pi}\Gamma(\alpha)\sin\left(\frac{\pi\alpha}{2}\right)$.

3.3 Tempered Stable Distributions

Mandelbrot (Mandelbrot, B. B. (1963). The variation of certain speculative prices, Journal of Business 36: 394–419.) proposed stable distributions for modeling in finance, but over the years his hypothesis lost significance in comparison to competing models. In the mid 1990s the stable distribution hypothesis, once dismissed, returned to the forefront of scientific discourse. Cont, Potters, and Bouchaud (Cont, R., Potters, M. and Bouchaud, J.-P. (1997). Scaling in stock market data: Stable laws and beyond, in B. Dubrulle, F. Graner, D. Sornette (eds.) Scale Invariance and Beyond, Proceedings of the CNRS Workshop on Scale Invariance, Springer, Berlin.) found strong empirical evidence that high-frequency returns align well with a stable distribution, extending as far as six standard deviations from the mean. In the case of more extreme returns, the distribution exhibited an approximately exponential decay. To account for these facts, Mantegna and Stanley (Mantegna, R. N. and Stanley, H. E. (1994).

Stochastic processes with ultraslow convergence to a Gaussian: The truncated Levy flight, ´ Physical Review Letters 73: 2946–2949.) introduced the truncated Lévy distributions (TLD) with sharp truncation of the stable PDF at some point. In contrary, Koponen (Koponen, I. (1995). Analytic approach to the problem of convergence of truncated Levy flights towards the Gaussian stochastic process, Physical Review E 52: 1197–1199.) proposed exponential smoothing which leads to the following characteristic function:

$$\ln(\varphi(\xi)) = -\frac{\gamma^\alpha}{\cos\left(\frac{\pi\alpha}{2}\right)}\left(\left(\xi^2+\lambda^2\right)^{\alpha/2}\cos\left(\alpha\arctan\left(\frac{|\xi|}{\lambda}\right)\right)-\lambda^\alpha\right), \tag{3.5}$$

where $\alpha \neq 1$ is the tail exponent, γ is the scale parameter and λ is the truncation coefficient (for simplicity β and δ are set to zero). Clearly, the symmetric TLD reduces to the symmetric stable distribution ($\beta = \delta = 0$) when $\lambda = 0$. For small to moderate returns, the TLD closely mimics a stable distribution. However, for extreme returns, the truncation leads the distribution to converge toward a Gaussian, ensuring that all moments are finite. This dual behavior implies that modeling asset returns with a TLD captures both the short-term heavy-tailed characteristics and the long-run tendency toward normality.
The (exponentially smoothed) TLD was not much recognized in finance until the introduction of the KoBoL (Boyarchenko and Levendorskii, Boyarchenko, S. I. and Levendorskii, S. Z. (2000). Option pricing for truncated Levy processes, ´ International Journal of Theoretical and Applied Finance 3: 549–552.) and CGMY models (Carr et al., Carr, P., Geman, H., Madan, D. B. and Yor, M. (2002). The fine structure of asset returns: an empirical investigation, Journal of Business 75: 305–332.). The term **Tempered Stable Distribution** (TSD), by which Koponen's TLD is currently known in mathematical contexts, was coined by Rosiński around this time (Rosiński, Rosinski, J. (2007). Tempering stable processes, Stochastic Processes and Their Applications 117(6): 677–707.). Despite their appealing statistical properties, tempered stable distributions have seen limited application thus far. This is likely due to their complex formulation: as with stable distributions, only the characteristic function is available, while closed-form expressions for the density and distribution functions remain unknown.

Exercises

3. Plot the PDFs and CDFs of the three analytically known standard stable distributions from Section 3.1.

4. Implement the direct integration method from Section 3.4 in order to approximate the PDF of $X \sim \mathcal{S}_0^\alpha(1, \beta, 0)$. Further, derive the PDF of $Y \sim \mathcal{S}_0^\alpha(3, \beta, 2)$. Plot both PDFs for different values of α, β.

4 Renewal Theory

In the most general form, we concern ourselves with **renewal theory**, a branch of probability theory that generalizes the Poisson process by allowing for arbitrary holding times. Let $(T_n)_{n\in\mathbb{N}}$ denote the n-th jump times and consider the **renewal intervals** $([T_{n-1}, T_n])_{n\in\mathbb{N}}$. We denote the Lebesgue measure with λ and the Lebesgue measure of an interval is just the length of that interval. Hence, the n-th holding times can be written as

$$(\Delta T_n)_{n\in\mathbb{N}} = (\lambda\left([T_{n-1}, T_n]\right))_{n\in\mathbb{N}} = (T_n - T_{n-1})_{n\in\mathbb{N}}. \tag{4.1}$$

Define the indicator function

$$\mathbb{1}_{\{T_n \le t\}} = \begin{cases} 1, & \text{if } T_n \le t, \\ 0, & \text{otherwise.} \end{cases} \tag{4.2}$$

Now the number of jumps that have occurred at time $t \ge 0$, i.e.

$$N_t := \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \le t\}} = \sup\{n \in \mathbb{N} : T_n \le t\}, \tag{4.3}$$

form the stochastic **renewal process** $N(t)_{t\ge 0}$.
Let the **rewards** $(S_k)_{k\in\mathbb{N}}$ be a sequence of i.i.d. random variables satisfying $\mathbb{E}[|S_1|] < \infty$. Then the random variable

$$(Y_t)_{t\ge 0} := \left(\sum_{k=1}^{N_t} S_k\right)_{t\ge 0} \tag{4.4}$$

is called a **renewal-reward process**. It is not any more necessary that the N_j and S_k are independent for every $j, k \in \mathbb{N}$ which is one of the strengths of this general theory. Due to the elementary renewal theorems, the **renewal function** $m(t) := \mathbb{E}[N_t]$ satisfies asymptotically

$$\lim_{t\to\infty} \frac{m(t)}{t} = \frac{1}{\mathbb{E}[\Delta T_1]}. \tag{4.5}$$

Further, for the **reward function** $g(t) := \mathbb{E}[Y_t]$, we have the asymptotics

$$\lim_{t\to\infty} \frac{g(t)}{t} = \frac{\mathbb{E}[S_1]}{\mathbb{E}[\Delta T_1]}. \tag{4.6}$$

In addition, the renewal function satisfies the **renewal equation**

$$m(t) = F_{\Delta T_1}(t) + \int_0^t m(t-s) \cdot f_{\Delta T_1}(s)\,\mathrm{d}s \tag{4.7}$$

with the CDF $F_{\Delta T_1}$ and the PDF $f_{\Delta T_1}$ of ΔT_1. Let $(N_t)_{t\ge 0}$ be a renewal process with renewal function $m(t) = \mathbb{E}[N_t]$ and inter-renewal mean μ. Let $g : [0,\infty) \to [0,\infty)$ be a monotone and non-increasing function satisfying

$$\int_0^{\infty} g(t)\,\mathrm{d}t < \infty. \tag{4.8}$$

Then the **key renewal theorem** states, that

$$\lim_{t\to\infty}\int_0^t g(t-x)m'(x)\,\mathrm{d}x = \frac{1}{\mu}\int_0^\infty g(x)\,\mathrm{d}x. \tag{4.9}$$

5 Simulation of Stable Random Variables

5.1 Sampling Algorithms for $\mathcal{N}(0,1) = \mathsf{S}^2(\beta,1,0)$

The Box-Muller Algorithm

For simulating standard normal distributed random variables X and Y, we follow the Box-Muller algorithm (Janicki and Weron, 1994b):

- Generate two independent random variables U, V uniformly distributed on the unit interval $(0,1)$;
- Let
$$X = \sqrt{-2\ln(U)}\cos(2\pi V), \tag{5.1}$$
and
$$Y = \sqrt{-2\ln(U)}\sin(2\pi V), \tag{5.2}$$

Then X and Y are independent random variables with a standard normal distribution.

The Marsaglia Polar Method

This algorithm is a modification of the Box-Muller algorithm which does not require computation of the sine and cosine functions:

- Generate two independent random variables $U, V \sim \mathsf{Unif}((-1,1))$ and calculate
$$S = U^2 + V^2. \tag{5.3}$$
- If $S \geq 1$, start the algorithm from the beginning.
- Let
$$X = U\sqrt{\frac{-2\ln(S)}{S}} \tag{5.4}$$
and
$$Y = V\sqrt{\frac{-2\ln(S)}{S}}. \tag{5.5}$$

Then X and Y are independent random variables with a standard normal distribution.

The Inverse Transform Sampling

We can also generate standard normal i.i.d. random variables via the Inverse of the its CDF:

- Generate a random variable $U \sim \mathsf{Unif}((0,1))$.
- Then,
$$X = \Phi^{-1}(U) \sim \mathcal{N}(0,1). \tag{5.6}$$
Here $\Phi(x)$ is the CDF of the standard normal distribution $\mathcal{N}(0,1)$.

5.2 Sampling Algorithm for $\text{Lévy}(\gamma,\delta) = \mathsf{S}^{1/2}(1,\gamma,\delta)$

Random samples from the Lévy distribution can be generated using inverse transform sampling. Given a random variable $U \sim \mathcal{U}((0,1])$ drawn from the uniform distribution on the unit interval $(0,1]$, the random variable X given by

$$X = F^{-1}(U) = \frac{\gamma}{(\Phi^{-1}(1-U/2))^2} + \delta$$

is Lévy-distributed $\mathsf{Lévy}(\gamma,\delta)$ with scale γ and location δ. Here $\Phi(x)$ is the CDF of the standard normal distribution $\mathcal{N}(0,1)$.

5.3 The Chambers-Mallows-Stuck Algorithm for $\mathsf{S}_0^{\alpha}(\beta,\gamma,\delta)$

Sampling stable random variables is more complicated, since there exist no explicit analytic Form of for the inverse $F^{-1}(x)$ nor the CDF $F(x)$ itself. Hence, techniques like rejection or inversion methods do not work. The CMS algorithm by Chambers, Mallows and Struck (1976) allows you to easily sample independent stable random variables. For a detailed proof, see Weron (1996).

- Generate $U \sim \mathsf{Unif}((-\frac{\pi}{2},\frac{\pi}{2}))$, i.e. a random variable uniformly distributed on $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ and $V \sim \mathsf{Exp}(1)$ an independent exponential random variable W with mean 1. Here, $\mathsf{Exp}(1) = \mathsf{Exp}(\lambda = 1) = \mathsf{Exp}(\frac{1}{\mu})$ with rate λ being the reciprocal of the mean μ.
- Define
$$\xi = \begin{cases} \frac{1}{\alpha}\arctan(-\zeta), & \alpha \neq 1, \\ \frac{\pi}{2}, & \alpha = 1, \end{cases} \tag{5.7}$$
and
$$\zeta = -\beta \tan\left(\frac{\pi\alpha}{2}\right). \tag{5.8}$$
- For $\alpha \neq 1$ compute:
$$X = (1+\zeta^2)^{\frac{1}{2\alpha}} \frac{\sin(\alpha(U+\xi))}{\cos(U)^{1/\alpha}} \left[\frac{\cos(U-\alpha(U+\xi))}{V}\right]^{\frac{1-\alpha}{\alpha}}, \tag{5.9}$$

- For $\alpha = 1$ compute:

$$X = \frac{1}{\xi}\left(\left(\frac{\pi}{2} + \beta U\right)\tan(U) - \beta \ln\left(\frac{\frac{\pi}{2} V \cos(U)}{\frac{\pi}{2} + \beta U}\right)\right), \tag{5.10}$$

- This yields $X \sim \mathsf{S}_0^\alpha(\beta, 1, 0)$, i.e. a standard α-stable distributed random variable. Now we compute a random variable $Y \sim \mathsf{S}_0^\alpha(\beta, \gamma, \delta)$ via

$$Y = \begin{cases} \gamma X + \delta, & \alpha \neq 1, \\ \gamma X + \frac{2}{\pi}\beta\gamma \ln(\gamma) + \delta, & \alpha = 1. \end{cases} \tag{5.11}$$

  Now, $Y \sim \mathsf{S}_0^\alpha(\beta, \gamma, \delta)$.

For $\alpha = 2$, the standard CMS algorithm reduces to the Box-Muller algorithm for sampling standard Gaussian random variables from Section 4.1. According to Misiorek & Weron (2010), the CMS algorithm is regarded as the fastest and most accurate at that time.

Exercises

5. Implement the Box-Muller algorithm from Section 4.1!
6. Implement the Marsaglia Polar Method from Section 4.1!
7. Implement the Inverse Transform Sampling methods from Section 4.1 & 4.2!
8. Implement the Chambers-Mallows-Stuck algorithm from Section 4.3!

6 Estimation of (Tempered) Stable Parameters

6.1 Approximation of PDFs and CDFs from CFs

The lack of closed form formulas for most (tempered) stable densities and distribution functions has negative consequences. How can we approximate these objects? We present two methods: A **direct numerical integration** method for stable distributions and some **fast Fourier transform** (FTT) approach for the tempered and genuine stable distributions.

Direct Numerical Integration

In the **direct numerical integration method**, explicit formulas for the PDF and CDF of the standard stable distributions are computed using numerical integration. The following integral representations rely on Zolotarev (1986). For proofs and properties of the involved functions, we refer to Nolan's monograph, Nolan (2020).

We begin with a representation for the standard stable PDF of $\mathsf{S}_0^\alpha(\beta,1,0)$. Let

$$\eta := \begin{cases} \frac{1}{\alpha}\arctan\left(\beta\tan\left(\frac{\pi\alpha}{2}\right)\right), & \alpha \neq 1, \\ \frac{\pi}{2}, & \alpha = 1. \end{cases} \tag{6.1}$$

Furthermore, define the function

$$V(\vartheta;\alpha,\beta) = \begin{cases} (\cos(\alpha\eta))^{\frac{1}{\alpha-1}}\left(\frac{\cos(\vartheta)}{\sin(\alpha(\eta+\vartheta))}\right)^{\frac{\alpha}{\alpha-1}}\frac{\cos(\alpha\eta+(\alpha-1)\vartheta)}{\cos(\vartheta)}, & \alpha \neq 1, \\ \frac{2}{\pi}\left(\frac{\frac{\pi}{2}+\beta\vartheta}{\cos(\vartheta)}\right)\exp\left(\frac{1}{\beta}\left(\frac{\pi}{2}+\beta\vartheta\right)\tan(\vartheta)\right), & \alpha = 1,\ \beta \neq 0. \end{cases} \tag{6.2}$$

Then, for $\alpha \neq 1$, the standard stable PDF is given by

$$f(x;\alpha,\beta) = \begin{cases} \frac{\alpha x^{\frac{1}{\alpha-1}}}{\pi(\alpha-1)}\int_{-\eta}^{\frac{\pi}{2}} V(\vartheta;\alpha,\beta)\exp\left(-x^{\frac{\alpha}{\alpha-1}}V(\vartheta;\alpha,\beta)\right)\,\mathrm{d}\vartheta, & x > 0, \\ \Gamma\left(1+\frac{1}{\alpha}\right)\cos(\eta)\cos(\alpha\eta)^{\frac{1}{\alpha}}/\pi, & x = 0, \\ f(-x,\alpha,-\beta), & x < 0. \end{cases} \tag{6.3}$$

In case of $\alpha = 1$, we get

$$f(x;1,\beta) = \begin{cases} \frac{1}{2|\beta|}e^{\frac{\pi x}{2\beta}}\int_{-\pi/2}^{\pi/2} V(\vartheta;1,\beta)\exp\left(-e^{\frac{\pi x}{2\beta}}V(\vartheta;1,\beta)\right)d\vartheta, & \beta \neq 0, \\ \frac{1}{\pi(1+x^2)}, & \beta = 0. \end{cases} \tag{6.4}$$

Now, we present representations of the CDF of the standard stable PDF of $\mathsf{S}_0^\alpha(\beta,1,0)$. For $\alpha \neq 1$ is given by

$$F(x;\alpha,\beta) = \begin{cases} c(\alpha,\beta) + \frac{\mathrm{sign}(1-\alpha)}{\pi}\int_{-\eta}^{\frac{\pi}{2}}\exp\left(-x^{\frac{\alpha}{\alpha-1}}V(\vartheta;\alpha,\beta)\right)\,\mathrm{d}\vartheta, & x > 0, \\ \left(\frac{\pi}{2}-\eta\right)/\pi, & x = 0, \\ 1 - F(-x;\alpha,-\beta), & x < 0, \end{cases} \tag{6.5}$$

where

$$c(\alpha,\beta) = \begin{cases} \left(\frac{\pi}{2} - \eta\right)/\pi, & 0 < \alpha < 1, \\ 1, & 1 < \alpha < 2. \end{cases} \tag{6.6}$$

Finally, for $\alpha = 1$, we get

$$F(x;1,\beta) = \begin{cases} \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \exp\left(-e^{-\frac{\pi x}{2\beta}} V(\vartheta;1,\beta)\right) \mathrm{d}\vartheta, & \beta > 0, \\ \frac{1}{2} + \frac{1}{\pi}\arctan(x), & \beta = 0, \\ 1 - F(-x;\alpha,-\beta), & \beta < 0. \end{cases} \tag{6.7}$$

FFT-Approximation of Stable PDFs

In this method, we apply the Fourier inversion formula to the characteristic function in order to approximate the probability density function f of a stable distribution $\mathsf{S}(\alpha,\beta,\gamma,\delta;1)$. Here, we use the 1-parametrization. Due to the convergence of the integral, the relationship

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ix\xi}\cdot\varphi(\xi)\,\mathrm{d}\xi \approx \int_{-K}^{K} e^{-ix\xi}\cdot\varphi(\xi)\,\mathrm{d}\xi \tag{6.8}$$

holds for some large $K > 0$. Further, we want to evaluate the above integral for a big number N of equally-spaced points with small distance h. We use the resulting grid

$$((x_k))_{k=1,\dots,N} = \left(\left(k - 1 - \frac{N}{2}\right)h\right)_{k=1,\dots,N}.$$

- Chenyao, D., Doganoglu, T., Mittnik, S., 1998. An approximation procedure for asymmetric stable Paretian densities. Mathematical and Computer Modelling 29 (1999) 235-240. Statist. 13, 463–475.

  FFT approach with the left point rule (LPA). Yields $f_{LPA}(x)$.

- Menn, C. & Rachev, S. T. (2006). Calibrated FFT-based density approximations for alpha-stable distributions. Computational Statistics and Data Analysis, 50, 1891–1904.

  FFT approach with the midpoint rule (MPA). Yields $f_{MPA}(x)$.

- Simpson rule: $f(x) = \frac{1}{3}f_{LPA}(x) + \frac{2}{3}f_{MPA}(x)$

  OR:

  Exact weights via

  $$f_{Nolan}(x_k;\alpha,\beta) = \lambda_{\alpha,\beta}(x_k)f_{LPA}(x_k;\alpha,\beta) + (1-\lambda_{\alpha,\beta}(x_k))f_{MPA}(x_k;\alpha,\beta), \tag{6.9}$$

  for every $k = 1,\dots,N$.

Topics: FFT of CF, solving the Fourier integral via the Simpson rule, adjust the tails via Bergström's series expansions.

Comparison and Usage of the Direct and the FFT Method

As for the computational speed, the FFT-based approach is faster for large samples, whereas the direct integration method favors small data sets since it can be computed at any arbitrarily chosen point. Mittnik, Doganoglu and Chenyao (1999) report that for $N = 2^{13}$, the FFT-based method is faster for samples exceeding 100 observations and slower for smaller data sets. We must stress, however, that the FFT-based approach is not as universal as the direct integration method – it is efficient only for large alpha's and only as far as the PDF calculations are concerned. When computing the CDF, the former method must numerically integrate the density, whereas the latter takes the same amount of time in both cases.
To the best of our knowledge, currently no statistical computing environment offers the computation of stable density and distribution functions in its standard release. Users have to rely on third-party libraries or commercial products. A few are worth mentioning. The standalone program **STABLE** (downloadable from John Nolan's web page: `http://academic2.american.edu/~jpnolan/stable/stable.html`) is probably the most efficient. It was written in Fortran and calls several external IMSL routines, see Nolan (1997) for details.
Apart from speed, the STABLE program also exhibits high relative accuracy (ca. 10^{-13}; for default tolerance settings) for extreme tail events and 10^{-10} for values used in typical financial applications (like approximating asset return distributions). The STABLE program is also available in library form through Robust Analysis Inc. (`http://www.robustanalysis.com`). This library provides interfaces to Matlab, S-plus/R and Mathematica.
In the late 1990s, Diethelm Wurtz initiated the development of **Rmetrics**, an open source collection of S-plus/R software packages for computational finance (`http://www.rmetrics.org`). In the `fBasics` package, stable PDF and CDF calculations are performed using the direct integration method, with the integrals being computed by R's function `integrate`.
The FFT-based approach is utilized in **Cognity**, a commercial risk management platform that offers derivatives pricing and portfolio optimization based on the assumption of stably distributed returns (`http://www.finanalytica.com`). The FFT implementation is also available in Matlab (`stablepdf_fft.m`) from the Statistical Software Components repository (`https://ideas.repec.org/c/boc/bocode/m429004.html`).

Exercises

9. Implement the direct integration method for the stable PDF and CDF assuming $\alpha \in (1, 2)$.

10. Implement the direct integration method from Section 3.4 in order to approximate the PDF of $X \sim \mathcal{S}_0^\alpha(1, \beta, 0)$. Further, derive the PDF of

$Y \sim \mathcal{S}_0^{\alpha}(3, \beta, 2)$. Plot both PDFs for different values of α, β.